GGG-W-646
JULY 13, 1954
SUPERSEDING
Int. Fed. Spec. GGG-W-646 (Navy Ships)
8 May 1952

FEDERAL SPECIFICATION

WRENCH, OPEN END, RATCHET (TAC PATTERN), FOR TUBE FITTINGS, ELECTRIC CABLE TERMINALS AND STUFFING TUBE GLAND NUTS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** — This specification covers individual components of open-ratchet wrenches, and wrench sets, for tube fitting, stuffing gland nuts, and applications where difficulties arise in the use of standard-design sockets.

1.1.1 *Federal Specification coverage.*—This specification covers only the types, classes, styles, and forms of the commodity generally purchased by the Federal Government, and does not include all types, classes, styles, and forms which are commercially available.

1.2 **Classification.**—Individual components shall be of the following types, classes, styles, and forms, as specified (see 6.1):

Type I.—Wrench heads.

Class 1.—Open ratchet, crowfoot.
Style A.—Multipurpose.
Style B.—Single purpose.
Form A.—Hexagon.
Form B.—Open end.

Class 2.—Rigid, crowfoot.
Style A.—Multipurpose.

Type II.—Attachments.

Class 1.—Sockets, open detachable.
Style A.—Standard wall.
Form A.—Hexagon.
Form B.—Octagon.
Style B.—Thin wall.
Form A.—Hexagon.

Class 2.—Adapters.
Style A.—Square drive.
Style B.—Turnbuckle.

Class 3.—Internal wrenching.
Form A.—Hexagon.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

TT-E-489—Enamel; Gloss, Synthetic (for Exterior and Interior Surfaces).
TT-L-58 —Lacquer; Spraying, Clear and Pigmented (General Use).
GGG-W-641—Wrenches, Bolt and Nut; Socket.

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Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Specifications and standards applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 **Material.**—The material used shall be such as to produce wrenches conforming to the physical requirements specified herein.

3.2 **Finish.**

3.2.1 *Surface.*—All surfaces shall be substantially free from pits, nodules, burrs, cracks, and other detrimental defects. Surfaces usually ground, buffed, or otherwise finished and provided with a coating of chromium or nickel plate shall have a surface roughness equivalent to the finish produced by 180-grit or finer abrasive. Surfaces usually ground, buffed, or otherwise finished and provided with a coating as specified in 3.2.2, other than chromium or nickel plate, shall have a surface roughness equivalent to the finish produced by 80-grit or finer abrasive. Surfaces not usually ground or buffed shall have a surface finish conforming to the best manufacturing practices.

3.2.2 *Coatings.* — The coatings shall be adherent, smooth, continuous, and free from uncoated areas, pits, blisters, nodules, and any other defects which would interfere with their protective value and serviceability. Unless otherwise specified (see 6.1), wrenches and components shall have one or a combination of the coatings listed herein. If chromium and nickel-plate coatings are specified but are restricted by Government order one of the other coatings listed will be acceptable.

3.2.2.1 *Bright chromium plate.*—The finish shall be electrodeposited metals consisting of nickel, followed by chromium, the minimum thickness being 0.0002 inch for nickel and 0.00001 inch for chromium.

3.2.2.2 *Bright nickel plate.* — The finish shall be electrodeposited coating of nickel, and shall be not less than 0.0002 inch.

3.2.2.3 *Cadmium plate.*—The finish shall be electrodeposited coating of cadmium not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

3.2.2.4 *Zinc plate.* — The finish shall be electrodeposited coating of zinc not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

3.2.2.5 *Hot phosphating.*—The finish shall consist of a heavy chemically produced phosphate coating followed with a coating of preservative.

3.2.2.6 *Oxide coating.* — The finish shall consist of a heavy chemically produced oxide coating followed with a coating of preservative.

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3.2.2.7 *Enamel.*—The enamel coating shall be in accordance with Federal Specification TT-E-489.

3.2.2.8 *Lacquer.*—The lacquer coating shall be in accordance with Federal Specification TT-L-58.

3.3 **Marking.**—Wrench heads and attachments shall be stamped or otherwise permanently marked in a plain and legible manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.3.1 Type I, class 1, style A, multipurpose, open ratchet crowfoot wrench heads, and attachments type II, class 1, style A, forms A and B, when specified (see 6.1) shall be marked with code shown in tables I, V, and VI.

3.3.2 Type I, class 1, style B single-purpose wrench heads shall be marked with the size across the flats.

3.3.3 Type II, class 1, open detachable sockets shall, unless otherwise specified, be marked with the size across the flats. (See 6.1.)

3.3.4 Type II, class 1, style A, form B, octagon open detachable sockets shall, unless otherwise specified (see 6.1), be marked by code letters in accordance with table VI.

3.3.5 Type II, class 3, hexagon internal wrenching attachments shall be marked with the size across the flats.

3.4 **Hinged handles, slide rods, and universal joints.**—Hinged handles, slide rods and universal joints shall conform to Federal Specification GGG-W-641.

3.5 **Type I, wrench heads.**

3.5.1 *Class 1, open ratchet, crowfoot.*—Type I, class 1 wrench heads shall consist of a body, a slotted rotating member, and an enclosure plate for accessibility in repairs, and shall be reversible by inverting the wrench head.

3.5.1.1 *Style A, multipurpose.* — Type I, class 1, style A wrench heads shall be so designed that the end opposite the female square drive will have a rotating member to fit type II attachments of the intended size range. The rotating member shall have a spring-loaded ball located in the center in relation to the width of the head plus or minus 1/64 inch to aid the holding of sockets in the wrench head. The wrench heads shall conform to table I and be similar to figure 1.

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TABLE I.—*Type I, class 1, style A, open ratchet, crowfoot, multipurpose wrench heads*

Component number	Rated torque	Code	Dimensions							
			GG[1]	HH (Minimum)	JJ[1]	LL+.015 −.005	MM+.005 −.000	NN±.005	KK	
									Degrees	Minutes
	Inch-pounds		*Inches*		*Inches*	*Inch*	*Inch*	*Inches*		
1	500	----	1.380	0.520	1.820	0.430	0.254	0.690	7	30
2	1200	A-C	1.650	.645	2.210	.520	.381	.940	7	30
3	1500	D-F	2.150	.900	2.710	.644	.381	1.190	7	30
4	2250	G-M	2.750	1.400	3.470	.644	.508	1.720	7	30
5	3400	N-T	3.900	2.085	4.630	.710	.508	2.470	6	0
6	3750	V-Z	4.620	2.800	5.380	.770	.508	3.190	5	0

[1] Nominal dimension, held within drop-forging association tolerances.

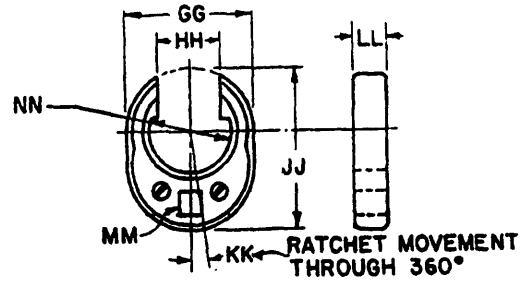


FIGURE 1.—Type I, class 1, style A, open ratchet, crowfoot, multipurpose wrench heads.

3.5.1.2 *Style B, single-purpose.*

3.5.1.2.1 *Form A, hexagon.*—Type I, class 1, style B, form A wrench heads shall have a broached hexagon opening in the rotating member, shall conform to table II and be similar to figure 2.

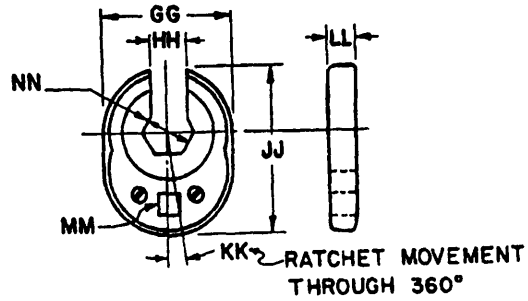


FIGURE 2.—Type I, class 1, style B, form A, open ratchet, crowfoot, single-purpose, hexagon wrench heads.

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TABLE II.—*Type I, class 1, style B, form A, open ratchet, crowfoot, single-purpose, hexagon wrench heads*

Component number	Rated torque	Hex nut size	Dimensions								
			NN		GG[1]	HH	JJ[1]	LL+.015 −.005	MM+.005 −.000	KK	
			Minimum	Maximum						Degrees	Minutes
	Inch-pounds	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*		
7	150	5/8	0.378	0.386	1.380	0.380+.015 −.000	1.820	0.322	0.254	7	30
8	180	7/16	.440	.448	1.380	.443+.015 −.000	1.820	.322	.254	7	30
9	200	1/2	.504	.512	1.380	.506+.015 −.000	1.820	.322	.254	7	30
10	225	9/16	.566	.575	1.630	.570+.015 −.000	2.200	.322	.381	7	30
11	380	5/8	.629	.638	1.630	.635+.015 −.000	2.200	.322	.381	7	30
12	420	11/16	.692	.701	1.933	.700+.015 −.000	2.433	.322	.381	7	30
13	460	3/4	.755	.765	2.053	.760+.015 −.000	2.593	.332	.381	7	30
14	520	13/16	.818	.828	2.173	.826+.015 −.000	2.733	.360	.381	7	30
15	750	7/8	.880	.890	2.301	.890+.015 −.000	2.901	.370	.381	7	30
16	820	15/16	.944	.995	2.421	.956+.015 −.000	3.021	.380	.381	7	30
17	1000	1	1.006	1.017	2.561	1.020+.015 −.000	3.161	.385	.381	7	30
176	1100	1 1/16	1.068	1.080	2.685	1.080+.015 −.000	3.285	.385	.381	7	30
177	1175	1 1/8	1.132	1.145	2.685	1.146+.015 −.000	3.285	.385	.381	7	30
178	1250	1 3/16	1.194	1.207	2.807	1.210+.015 −.000	3.407	.447	.381	7	30
18	1250	1 1/4	1.257	1.270	2.807	1.270+.015 −.000	3.407	.447	.381	7	30
179	1350	1 5/16	1.320	1.334	2.987	1.336+.015 −.000	3.687	.458	.508	7	30
19	1350	1 3/8	1.383	1.397	2.987	1.396+.015 −.000	3.687	.458	.508	7	30
180	1750	1 7/16	1.446	1.460	3.107	1.460+.015 −.000	3.807	.458	.508	7	30
20	1750	1 1/2	1.508	1.523	3.107	1.532+.020 −.000	3.807	.458	.508	7	30
181	1800	1 9/16	1.571	1.586	3.307	1.582+.020 −.000	4.007	.458	.508	7	30
21	1800	1 5/8	1.634	1.649	3.307	1.646+.020 −.000	4.007	.458	.508	7	30
182	1850	1 11/16	1.696	1.711	3.467	1.712+.020 −.000	4.167	.458	.508	6	0
22	1850	1 3/4	1.759	1.775	3.467	1.796+.020 −.000	4.167	.458	.508	6	0

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TABLE II.—*Type I, class 1, style B, form A, open ratchet, crowfoot, single-purpose, hexagon wrench heads (cont'd)*

Component number	Rated torque	Hex nut size	Dimensions								
			NN		GG[1]	HH	JJ[1]	LL+.015 −.005	MM+.005 −.000	KK	
			Minimum	Maximum						Degrees	Minutes
	Inch-pounds	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*		
183	1900	1 13/16	1.822	1.838	3.627	1.838+.020 −.000	4.327	0.458	0.508	6	0
23	1900	1 7/8	1.885	1.901	3.627	1.902+.020 −.000	4.327	.458	.508	6	0
184	2100	1 15/16	1.948	1.964	3.692	1.964+.020 −.000	4.392	.458	.508	6	0
24	2100	2	2.011	2.028	3.692	2.030+.020 −.000	4.392	.458	.508	6	0
185	2200	2 1/16	2.074	2.092	3.807	2.090+.020 −.000	4.527	.458	.508	6	0
25	2200	2 1/8	2.136	2.155	3.807	2.150+.020 −.000	4.527	.458	.508	6	0
186	3000	2 3/16	2.200	2.219	3.957	2.214+.020 −.000	4.707	.510	.508	5	0
26	3000	2 1/4	2.262	2.281	3.957	2.280+.020 −.000	4.707	.510	.508	5	0
187	3000	2 5/16	2.324	2.344	4.117	2.342+.020 −.000	4.867	.520	.508	5	0
27	3000	2 3/8	2.388	2.408	4.117	2.406+.020 −.000	4.867	.520	.508	5	0
188	3000	2 7/16	2.453	2.470	4.277	2.466+.020 −.000	5.027	.520	.508	5	0
28	3000	2 1/2	2.513	2.534	4.277	2.530+.020 −.000	5.027	.520	.508	5	0
189	3000	2 9/16	2.576	2.597	4.440	2.592+.020 −.000	5.190	.520	.508	5	0
29	3000	2 5/8	2.639	2.660	4.440	2.660+.020 −.000	5.190	.520	.508	5	0
190	3000	2 11/16	2.701	2.722	4.607	2.706+.020 −.000	5.367	.520	.508	5	0
30	3000	2 3/4	2.766	2.787	4.607	2.785+.020 −.000	5.367	.520	.508	5	0
191	3000	2 13/16	2.827	2.849	4.780	2.846+.020 −.000	5.540	.520	.508	5	0
31	2600	2 7/8	2.890	2.912	4.607	2.970+.020 −.000	5.367	.520	.508	5	0
192	3000	2 7/8	2.890	2.912	4.780	2.970+.020 −.000	5.540	.520	.508	5	0
193	3050	2 15/16	2.954	2.977	4.957	2.972+.020 −.000	5.737	.520	.508	5	0
32	3050	3	3.016	3.039	4.957	3.036+.020 −.000	5.737	.520	.508	5	0

[1] Nominal dimension, held within drop-forging association tolerances.

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TABLE III.—*Type I, class 1, style B, form B, open ratchet, crowfoot, single-purpose, open-end wrench heads*

Component number	Rated torque	Nominal wrench size	Dimensions							
			GG[1]	NN		JJ[1]	LL+.015 −.005	MM+.005 −.000	KK	
				Minimum	Maximum				Degrees	Minutes
	Inch-pounds	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*		
33	150	5/8	1.380	0.378	0.386	1.820	0.322	0.254	7	30
34	180	7/16	1.380	.440	.448	1.820	.322	.254	7	30
35	200	1/2	1.380	.504	.512	1.820	.322	.254	7	30
36	225	9/16	1.630	.566	.575	2.200	.322	.381	7	30
37	380	5/8	1.630	.629	.638	2.200	.322	.381	7	30
38	420	11/16	1.933	.692	.701	2.433	.322	.381	7	30
39	460	3/4	2.053	.755	.765	2.593	.332	.381	7	30
40	520	13/16	2.173	.818	.828	2.733	.360	.381	7	30
41	750	7/8	2.301	.880	.890	2.901	.370	.381	7	30
42	820	15/16	2.421	.944	.955	3.021	.380	.381	7	30
43	1000	1	2.561	1.006	1.017	3.161	.385	.381	7	30
194	1070	1 1/16	2.685	1.068	1.080	3.285	.385	.381	7	30
195	1155	1 1/8	2.685	1.132	1.145	3.285	.385	.381	7	30
196	1250	1 3/16	2.807	1.194	1.207	3.407	.447	.381	7	30
44	1250	1 1/4	2.807	1.257	1.270	3.407	.447	.381	7	30
197	1350	1 5/16	2.987	1.320	1.334	3.687	.458	.508	7	30
45	1350	1 3/8	2.987	1.383	1.397	3.687	.458	.508	7	30
198	1750	1 7/16	3.107	1.446	1.460	3.807	.458	.508	7	30
46	1750	1 1/2	3.107	1.508	1.523	3.807	.458	.508	7	30
199	1800	1 9/16	3.307	1.571	1.586	4.007	.458	.508	7	30
47	1800	1 5/8	3.307	1.634	1.649	4.007	.458	.508	7	30
200	1850	1 11/16	3.467	1.696	1.711	4.167	.458	.508	6	0
48	1850	1 3/4	3.467	1.759	1.775	4.167	.458	.508	6	0
201	1900	1 13/16	3.627	1.822	1.838	4.327	.458	.508	6	0
49	1900	1 7/8	3.627	1.885	1.901	4.327	.458	.508	6	0
202	2100	1 15/16	3.692	1.948	1.964	4.392	.458	.508	6	0
50	2100	2	3.692	2.011	2.028	4.392	.458	.508	6	0
203	2200	2 1/16	3.807	2.074	2.092	4.527	.458	.508	6	0
51	2200	2 1/8	3.807	2.136	2.155	4.527	.458	.508	6	0
204	3000	2 3/16	3.957	2.200	2.219	4.707	.510	.508	5	0
52	3000	2 1/4	3.957	2.262	2.281	4.707	.510	.508	5	0
205	3000	2 5/16	4.117	2.324	2.344	4.867	.520	.508	5	0
53	3000	2 3/8	4.117	2.388	2.408	4.867	.520	.508	5	0
206	3000	2 7/16	4.277	2.453	2.491	5.027	.520	.508	5	0
54	3000	2 1/2	4.277	2.513	2.534	5.027	.520	.508	5	0
207	3000	2 9/16	4.440	2.576	2.587	5.190	.520	.508	5	0
55	3000	2 5/8	4.440	2.639	2.660	5.190	.520	.508	5	0
208	3000	2 11/16	4.607	2.701	2.722	5.367	.520	.508	5	0
56	3000	2 3/4	4.607	2.766	2.787	5.367	.520	.508	5	0
209	3000	2 13/16	4.780	2.827	2.849	5.540	.520	.508	5	0
57	3000	2 7/8	4.780	2.890	2.912	5.540	.520	.508	5	0
210	3050	2 15/16	4.957	2.954	2.977	5.737	.520	.508	5	0
58	3050	3	4.957	3.016	3.039	5.737	.520	.508	5	0

[1] Nominal dimension, held within drop-forging association tolerances.

[1] style B, form A wrench heads shall have

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3.5.1.2.2 *Form B, open end.*—Type I, class 1, style B, form B wrench heads shall have an open-end wrench style opening in the rotating member, and shall conform to table III and be similar to figure 3.

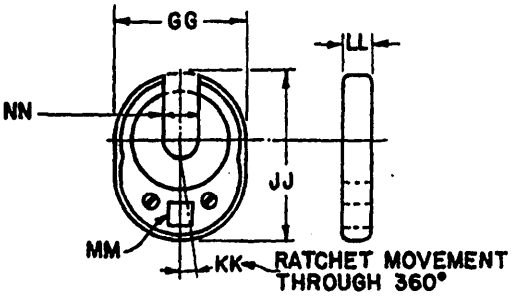


FIGURE 3.—Type I, class 1, style B, form B, open ratchet, crowfoot, single-purpose, open end wrench heads.

3.5.2 *Class 2, rigid, crowfoot, style A, multipurpose.* — Type I, class 2, style A wrench heads shall have a female square-drive socket at one end, and shall be so designed that the opposite end will fit type II attachments. Wrench heads shall conform to table IV and be similar to figure 4.

TABLE IV.—*Type I, class 2, style A, rigid, crowfoot, multipurpose wrench heads*

Component number	Rated torque	Dimensions					
		Opening A Minimum	Collar diameter B ± .005	Over-all width C +.032 −.032	Drive offset D ± .010	Thickness E +.032 −.032	Square drive F +.005 −.000
	Inch-pounds	*Inch*	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*
59	500	0.520	0.690	1.120	0.645	0.375	0.254
60	1200	.645	.940	1.420	.940	.500	.381
61	1500	.900	1.190	1.720	1.100	.500	.381

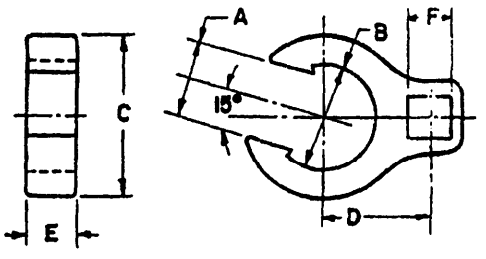


FIGURE 4.—Type I, class 2, style A, rigid, crowfoot, multipurpose wrench heads.

3.6 **Type II, attachments.**—Type II attachments shall be designed on one end to fit type I, class 1, style A, multipurpose wrench heads within the intended size range.

3.6.1 *Class 1, sockets, open detachable.*—Type II, class 1 sockets shall be slotted the entire length of the socket to allow the socket to slide over the cable or tubing and seat on the nut used with the corresponding cable or tubing size. Forged and machined sockets shall have a Rockwell hardness of 40 to 52 on the C scale. Cast sockets shall have a Rockwell hardness of 30 to 45 on the C scale.

3.6.1.1 *Style A, standard wall.*

3.6.1.1.1 *Form A, hexagon.*—Type II, class 1, style A, form A sockets shall conform to table V and be similar to figure 5.

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TABLE V.—*Type II, class 1, style A, form A, standard wall hexagon sockets*

Component number	Rated torque	Hex nut size	Code	Dimensions						
				AA		BB Minimum	CC±.032	DD Minimum	EE±.032	FF
				Minimum	Maximum					
	Inch-pounds			*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inches*	*Inches*
62	400	0.375	---	0.378	0.388	0.245	0.937	0.250	1.030	0.678+.005 −.005
63	420	.4375	---	.440	.450	.307	.937	.281	1.030	.678+.005 −.005
64	440	.500	---	.504	.514	.370	.937	.328	1.062	.678+.005 −.005
65	460	.5625	---	.566	.577	.370	1.000	.328	1.062	.678+.005 −.005
66	480	.625	---	.629	.641	.432	1.062	.328	1.125	.678+.005 −.005
67	500	.6875	---	.692	.704	.432	1.125	.343	1.125	.678+.005 −.005
68	720	.750	---	.755	.768	.557	1.125	.343	1.235	.926+.005 −.005
69	780	.8125	---	.818	.831	.557	1.187	.343	1.268	.926+.005 −.005
70	840	.875	---	.880	.894	.682	1.250	.500	1.298	.926+.005 −.005
71	920	.9375	---	.944	.959	.682	1.312	.562	1.330	.926+.005 −.005
72	980	1.000	A	1.006	1.021	.682	1.375	.562	1.360	.926+.005 −.005
73	1040	1.0625	---	1.068	1.083	.682	1.437	.593	1.392	.926+.005 −.005
74	1100	1.125	B	1.132	1.148	.682	1.500	.593	1.392	.926+.005 −.005
75	1200	1.125	---	1.132	1.148	.651	1.562	.593	1.875	.926+.005 −.005
76	1175	1.125	---	1.132	1.172	.812	1.834	.593	1.460	1.182+.000 −.015
77	1250	1.187	---	1.206	1.248	.812	1.834	.593	1.460	1.182+.000 −.015
78	1350	1.250	C	1.270	1.312	.812	1.853	.593	1.460	1.182+.000 −.015
79	1425	1.3125	---	1.332	1.374	.875	1.840	.718	1.580	1.182+.000 −.015
80	1500	1.375	DE	1.395	1.437	.875	1.912	.718	1.580	1.182+.000 −.015
81	1580	1.4375	F	1.458	1.500	.906	1.985	.718	1.580	1.182+.000 −.015
82	1500	1.250	---	1.270	1.312	.812	2.160	.593	1.46	1.710+.000 −.015
83	1500	1.3125	---	1.332	1.374	.875	2.150	.718	1.580	1.710+.000 −.015
84	1500	1.375	---	1.395	1.437	.875	2.200	.718	1.580	1.710+.000 −.015
85	1575	1.4375	---	1.458	1.500	1.031	2.260	.718	1.580	1.710+.000 −.015

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TABLE V.—*Type II, class 1, style A, form A, standard wall hexagon sockets (cont'd)*

Component number	Rated torque	Hex nut size	Code	Dimensions						
				AA		BB Minimum	CC±.032	DD Minimum	EE±.032	FF
				Minimum	Maximum					
	Inch-pounds			*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
86	1650	1.500	---	1.521	1.563	1.062	2.270	0.780	1.650	1.710+.000 −.015
87	1730	1.5625	---	1.583	1.625	1.093	2.280	.780	1.650	1.710+.000 −.015
88	1795	1.625	G	1.646	1.688	1.125	2.290	.780	1.650	1.710+.000 −.015
89	1860	1.6875	---	1.708	1.750	1.156	2.350	.780	1.650	1.710+.000 −.015
90	1940	1.750	J	1.771	1.813	1.187	2.347	.780	1.650	1.710+.000 −.015
91	2025	1.8125	---	1.833	1.875	1.218	2.418	.780	1.650	1.710+.000 −.015
92	2100	1.875	K	1.896	1.938	1.250	2.491	.968	1.840	1.710+.000 −.015
93	2160	1.9375	L	1.958	2.000	1.281	2.563	.780	1.650	1.710+.000 −.015
94	2250	2.000	---	2.021	2.063	1.344	2.635	.780	1.650	1.710+.000 −.015
95	2350	2.125	M	2.146	2.188	1.406	2.768	.780	1.650	1.710+.000 −.015
96	2300	2.0625	---	2.083	2.125	1.437	3.020	.843	1.775	2.458+.000 −.015
97	2360	2.125	---	2.146	2.188	1.500	3.050	.843	1.775	2.458+.000 −.015
98	2430	2.1875	N	2.208	2.250	1.562	3.100	.843	1.775	2.458+.000 −.015
99	2500	2.250	---	2.271	2.313	1.625	2.924	.905	1.840	2.458+.000 −.015
100	2560	2.3125	P	2.334	2.376	1.625	2.997	.905	1.840	2.458+.000 −.015
101	2620	2.375	---	2.396	2.438	1.687	3.068	.905	1.840	2.458+.000 −.015
102	2680	2.4375	R	2.458	2.500	1.750	3.140	.905	1.840	2.458+.000 −.015
103	2750	2.500	---	2.521	2.563	1.812	3.213	1.030	1.960	2.458+.000 −.015
104	2830	2.5625	---	2.583	2.625	1.812	3.284	1.030	1.960	2.458+.000 −.015
105	2890	2.625	---	2.646	2.688	1.825	3.357	1.030	1.960	2.458+.000 −.015
106	2950	2.6875	---	2.708	2.750	1.937	3.428	1.030	1.960	2.458+.000 −.015
107	3010	2.750	---	2.771	2.813	2.000	3.501	1.280	2.210	2.458+.000 −.015
108	3080	2.8125	---	2.833	2.875	2.000	3.572	1.030	1.960	2.458+.000 −.015
109	3140	2.875	S	2.896	2.938	2.062	3.646	1.030	1.960	2.458+.000 −.015

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TABLE V.—*Type II, class 1, style A, form A, standard wall hexagon sockets (cont'd)*

Component number	Rated torque	Hex nut size	Code	Dimensions						
				AA		BB Minimum	CC±.032	DD Minimum	EE±.032	FF
				Minimum	Maximum					
	Inch-pounds			*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*
110	3200	2.9375	---	2.958	3.000	2.062	3.717	1.030	1.960	2.458+.000 −.015
111	3260	3.000	---	3.021	3.063	2.062	3.790	1.030	1.960	2.458+.000 −.015
112	3320	3.0625	T	3.083	3.125	2.062	3.862	1.030	1.960	2.458+.000 −.015
113	3400	3.125	---	3.146	3.188	2.062	3.934	1.030	1.960	2.458+.000 −.015
114	3320	3.0625	---	3.083	3.125	2.062	3.862	1.343	2.364	3.176+.000 −.015
115	3350	3.125	---	3.146	3.188	2.093	3.934	1.343	2.364	3.176+.000 −.015
116	3380	3.1875	---	3.209	3.251	2.218	4.007	1.343	2.364	3.176+.000 −.015
117	3410	3.250	---	3.271	3.313	2.218	4.079	1.343	2.364	3.176+.000 −.015
118	3440	3.3125	---	3.333	3.375	2.218	4.151	1.343	2.364	3.176+.000 −.015
119	3470	3.375	---	3.396	3.438	2.343	4.223	1.343	2.364	3.176+.000 +.015
120	3500	3.4375	---	3.458	3.500	2.343	4.295	1.343	2.364	3.176+.000 −.015
121	3530	3.500	---	3.521	3.563	2.343	4.368	1.343	2.364	3.176+.000 −.015
122	3560	3.5625	---	3.584	3.626	2.531	4.440	1.343	2.364	3.176+.000 −.015
123	3590	3.625	---	3.646	3.688	2.531	4.512	1.343	2.364	3.176+.000 −.015
124	3620	3.6875	---	3.709	3.751	2.640	4.584	1.343	2.364	3.176+.000 −.015
125	3650	3.750	---	3.771	3.813	2.640	4.656	1.343	2.364	3.176+.000 −.015
126	3680	3.8125	---	3.834	3.876	2.812	4.729	1.343	2.364	3.176+.000 −.015
127	3710	3.875	---	3.896	3.938	2.812	4.800	1.343	2.364	3.176+.000 −.015
128	3730	3.9375	---	3.958	4.000	2.812	4.873	1.343	2.364	3.176+.000 −.015
129	3750	4.000	---	4.021	4.063	2.812	4.945	1.343	2.364	3.176+.000 −.015

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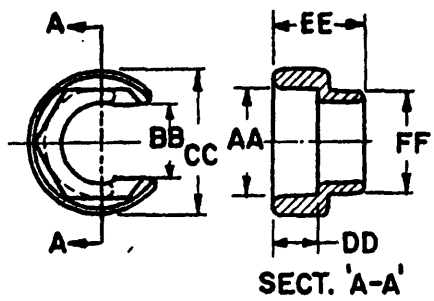


FIGURE 5.—Type II, class 1, style A, form A, standard wall hexagon sockets.

3.6.1.1.2 *Form B, octagon.*—Type II, class 1, style A, form B socket shall conform to table VI and be similar to figure 6.

TABLE VI.—*Type II, class 1, style A, form B, standard wall octagon sockets*

Component number	Rated torque	Gland nut octagon size	Code	Dimensions						
				AA[1]		BB Minimum	CC ±.032	DD Minimum	EE ±.032	FF +.000 −.015
				Minimum	Maximum					
	Inch-pounds	*Inches*		*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inches*	*Inches*
130	800	1.031	A	1.083	1.125	0.656	1.433	0.343	1.020	0.930
131	900	1.125	B	1.177	1.219	.656	1.534	.343	1.020	.930
132	1000	1.250	C	1.302	1.344	.656	1.670	.343	1.020	.930
133	1100	1.375	D-E	1.427	1.469	.906	1.805	.343	1.144	1.182
134	1150	1.437	F	1.490	1.532	.906	1.873	.343	1.144	1.182
135	1300	1.656	G	1.708	1.750	1.406	2.109	.343	1.175	1.710
136	1300	1.781	J	1.833	1.875	1.406	2.244	.343	1.175	1.710
137	1400	1.875	K	1.927	1.969	1.406	2.346	.343	1.175	1.710
138	1400	1.937	L	1.990	2.032	1.406	2.414	.343	1.175	1.710
139	1600	2.187	M	2.240	2.282	1.406	2.685	.343	1.175	1.710
140	1700	2.250	N	2.302	2.344	2.093	2.752	.343	1.241	2.458
141	1700	2.344	P	2.396	2.438	2.093	2.854	.343	1.241	2.458
142	1800	2.468	R	2.521	2.563	2.093	2.989	.343	1.241	2.458
143	2000	2.812	S	2.865	2.907	2.093	3.361	.343	1.241	2.458
144	2100	2.968	T	3.021	3.063	2.093	3.530	.343	1.241	2.458
145	2200	3.093	V	3.146	3.188	2.812	3.666	.343	1.301	3.176
146	2300	3.281	W	3.334	3.376	2.812	3.869	.343	1.301	3.176
147	2400	3.437	X	3.490	3.532	2.812	4.038	.343	1.301	3.176
148	2500	3.562	Y	3.615	3.657	2.812	4.173	.343	1.301	3.176
149	2700	3.750	Z	3.802	3.844	2.812	4.376	.343	1.301	3.176

[1] Oversize to allow for paint on packing and gland nuts.

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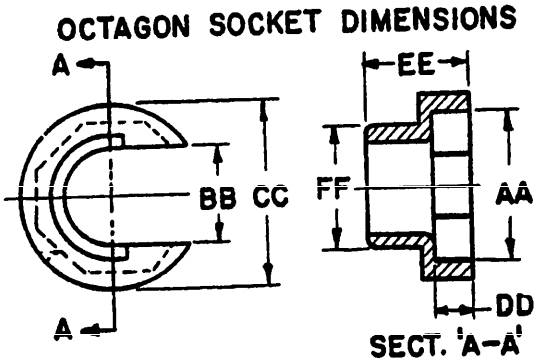


FIGURE 6.—Type II, class 1, style A, Form B, standard wall octagon sockets.

3.6.1.2 *Style B, thin wall, form A, hexagon.*—Type II, class 1, style B, form A sockets shall conform to table VII and be similar to figure 7.

TABLE VII.—*Type II, class 1, style B, form A, thin wall hexagon sockets*

Component number	Rated torque	Hex nut size	Dimensions						
			AA		BB Minimum	CC±.032	DD Minimum	EE±.032	FF±.005
			Minimum	Maximum					
	Inch-pounds	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inch*	*Inches*	*Inch*
150	200	0.375	0.378	0.388	0.245	0.750	0.250	1.030	0.678
151	200	.4375	.440	.450	.307	.750	.281	1.030	.678
152	240	.500	.504	.514	.370	.812	.328	1.062	.678
153	280	.5625	.566	.577	.370	.875	.328	1.062	.678
154	380	.625	.629	.641	.432	.937	.328	1.125	.678
155	480	.6875	.692	.704	.432	1.000	.343	1.125	.678

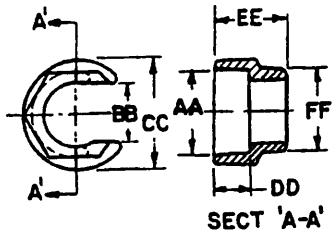


FIGURE 7.—Type II, class 1, style B, form A, thin wall hexagon sockets.

3.6.2 *Class 2, adapters.*

3.6.2.1 *Style A, square drive.*—Type II, class 2, style A adapters shall have a square drive tang opposite the end fitting the intended wrench head, shall conform to table VIII, and be similar to figure 8.

TABLE VIII.—*Type II, class 2, style A square drive adapters*

Component number	Body diameter N ±.005	Over-all length figure 9 P ±.032	Tang length R +.020 −.000	Square drive S +.000 −.005	Ball location U +.005 −.000
	Inches	*Inches*	*Inch*	*Inch*	*Inch*
156	0.678	0.777	0.280	0.252	0.140
157	.678	.877	.380	.377	.190
158	.926	.982	.380	.377	.190
159	.926	1.102	.500	.502	.250
160	1.175	1.216	.500	.502	.250

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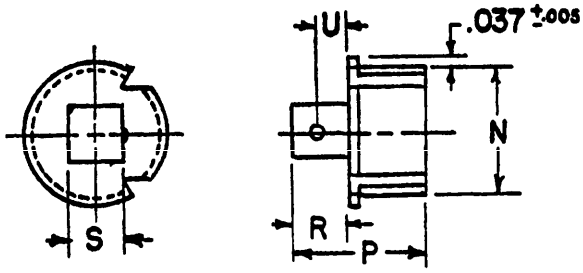


FIGURE 8.—Type II, class 2, style A, square drive adapters, flanged.

3.6.2.2 *Style B, turnbuckle.*—Type II, class 2, style B adapters shall consist of a collar and a drive pin conforming to table IX and be similar to figure 9. The collar shall be designed at one end to engage type I, class 1, style A multipurpose wrench head of the intended size; the opposite end shall slide over and partially encircle the turnbuckle, and shall have a hole drilled on center to accept the drive pin. The drive pin shall be designed to fit the hole in the collar extending through both collar walls, and shall have one end bent off center for ease of inserting and removing.

TABLE IX.—*Type II, class 2, style B, turnbuckle adapters*

Component number	Rated torque	Dimensions								
		G±.032	H+.015 −.000	J±.032	K±.032	L±.032	M±.032	N+.005 −.000	P±.032	R±.005
	Inch-pounds	*Inches*	*Inch*	*Inch*	*Inches*	*Inch*	*Inch*	*Inch*	*Inches*	*Inches*
161	150	0.875	0.531	0.625	0.980	0.550	0.218	0.128	1.250	0.678
162	250	1.250	.890	.845	1.265	.621	.250	.128	1.500	1.175

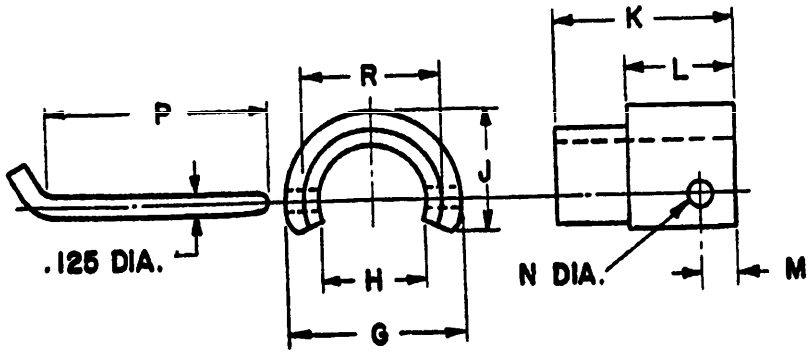


FIGURE 9.—Type II, class 2, style B, turnbuckle adapters.

3.6.3 *Class 3, internal wrenching attachment, form A, hexagon drive.*—Type II, class 3, form A attachment shall have a hexagon drive tang on one end, and the opposite end shall be designed to fit type I, class 1, style A (multipurpose) wrench heads. Attachments shall conform to table X and be similar to figure 10.

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TABLE X.—*Type II, class 3, form A, internal wrenching attachments*

Component number	Hexagon drive (V)		Body diameter (N) ±.005	Over-all length (T) +.032 −.032	Drive tang length (U) +.032 −.032
	Maximum	Minimum			
	Inches	*Inch*	*Inches*	*Inches*	*Inch*
163	0.1875	0.1860	0.678	0.815	0.328
164	.2187	.2172	.678	.815	.328
165	.250	.2485	.678	.815	.328
166	.3125	.3110	.678	.815	.328
167	.375	.3735	.678	.815	.328
168	.375	.3735	.926	1.000	.408
169	.500	.4985	.926	1.031	.439
170	.5625	.5600	.926	1.092	.500
171	.625	.6225	.926	1.218	.625
172	.5625	.5600	1.175	1.331	.625
173	.625	.6225	1.175	1.331	.625
174	.750	.7475	1.175	1.456	.750
175	1.000	.997	1.175	1.456	.750

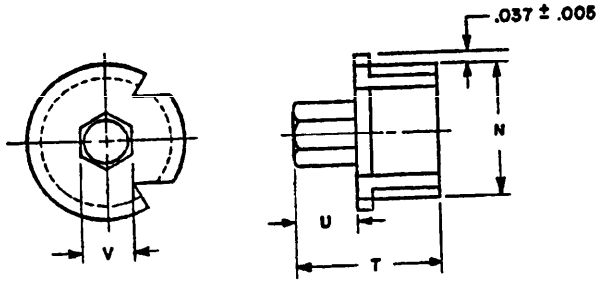


FIGURE 10.—Type II, class 3, form A, internal wrenching attachments, flanged.

3.7 **Metal box containers.**—Wrench sets shall be compactly housed in metal box containers. The containers shall have provision for holding the components securely in place. The containers shall be thoroughly cleaned, then painted inside and out with one coat of zinc-chromate primer, and finish-coated with light gray paint or enamel. Whenever the combined weight of the container and the components is 8 or more pounds, the container shall be furnished with handle or handles.

3.8 **Sets.**—Wrench sets shall contain the following components:

Set No. 1 *Quantity*	*Component No.*	*Nomenclature*
1	1	Multipurpose, open ratchet crowfoot wrench head.
1	62	Hexagon open detachable standard wall socket.
1	63	
1	64	
1	65	
1	66	
1	67	
1	---	¼-inch square drive hinged handle (see 3.4).
1	---	Metal box container (see 3.7).

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Set No. 2 Quantity	*Component No.*	*Nomenclature*
1	1	Multipurpose, open ratchet crowfoot wrench head.
1	150	Hexagon open detachable standard wall socket.
1	151	
1	152	
1	153	
1	154	
1	155	
1	---	¼-inch square drive hinged handle (see 3.4).
1	---	Metal box container (see 3.7).

Set No. 3 Quantity	*Component No.*	*Nomenclature*
1	2	Multipurpose, open ratchet crowfoot wrench head.
1	68	Hexagon open detachable standard wall socket.
1	69	
1	70	
1	71	
1	72	
1	73	
1	60	Multipurpose, rigid crowfoot wrench head.
1	158	Square drive adapter.
1	---	⅜-inch square drive hinged handle (see 3.4).
1	---	Metal box container (see 3.7).

Set No. 4 Quantity	*Component No.*	*Nomenclature*
1	3	Multipurpose, open ratchet crowfoot wrench head.
1	76	Hexagon open detachable standard wall socket.
1	78	
1	79	
1	80	
1	61	Multipurpose, rigid crowfoot wrench head.
1	160	Square drive adapter.
1	---	⅜-inch square drive hinged handle (see 3.4).
1	---	Metal box container (see 3.7).

Set No. 5 Quantity	*Component No.*	*Nomenclature*
1	4	Multipurpose, open ratchet crowfoot wrench head.
1	86	Hexagon open detachable standard wall socket.
1	90	
1	92	
1	94	
1	---	½-inch square drive hinged handle (see 3.4).
1	---	Metal box container (see 3.7).

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Set No. 6 *Quantity*	*Component No.*	*Nomenclature*
1	2	Multipurpose, open detachable crowfoot wrench head.
1	3	
1	4	
1	5	
1	130	Octagon open detachable standard wall socket.
1	131	
1	132	
1	133	
1	134	
1	135	
1	136	
1	137	
1	138	
1	139	
1	140	
1	141	
1	142	
1	143	
1	144	
1	---	3/8-inch square drive hinged handle (see 3.4).
1	---	1/2-inch square drive hinged handle (see 3.4).
1	---	3/8-inch universal joint.
1	---	1/2-inch universal joint.
1	---	Metal box container (see 3.7).

Set No. 7 *Quantity*	*Component No.*	*Nomenclature*
1	2	Multipurpose, open ratchet crowfoot wrench head.
1	3	
1	4	
1	5	
1	6	
1	130	Octagon open detachable standard wall socket.
1	131	
1	132	
1	133	
1	134	
1	135	
1	136	
1	137	
1	138	
1	139	
1	140	
1	141	
1	142	
1	143	
1	144	

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Set No. 7 (cont'd) Quantity	Component No.	Nomenclature
1	145	Octagon open detachable, standard wall socket.
1	146	
1	147	
1	148	
1	149	
1	---	3/8-inch square drive hinged handle (see 3.4).
1	---	1/2-inch square drive hinged handle (see 3.4).
1	---	3/8-inch universal joint (see 3.4).
1	---	1/2-inch universal joint (see 3.4).
1	---	Metal box container (see 3.7).

Set No. 8 Quantity	Component No.	Nomenclature
1	1	Multipurpose, open detachable crowfoot wrench head.
1	2	
1	3	
1	4	
1	5	
1	62	Hexagon open detachable standard wall socket.
1	63	
1	64	
1	65	
1	66	
1	67	
1	68	
1	69	
1	70	
1	71	
1	72	
1	73	
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1	98	

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Set No. 8 (cont'd) Quantity	Component No.	Nomenclature
1	99	Hexagon, open detachable standard wall socket.
1	100	
1	101	
1	102	
1	103	
1	104	
1	105	
1	106	
1	107	
1	108	
1	109	
1	110	
1	111	
1	112	
1	113	
1	59	Multipurpose rigid hand crowfoot wrench head.
1	60	
1	156	Square drive adapter.
1	157	
1	158	
1	159	
1	160	
1	161	Turnbuckle adapter.
1	162	
1	163	Hexagonal internal wrench.
1	164	
1	165	
1	166	
1	167	
1	169	
1	170	
1	172	
1	173	
1	---	1/4-inch square drive hinged handle (see 3.4).
1	---	3/8-inch square drive hinged handle (see 3.4).
1	---	1/2-inch square drive hinged handle (see 3.4).
1	---	Metal box container (see 3.7).

3.9 **Workmanship.** — The workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Inspection lot.**—All components of the same size, type, class, style, and form offered for delivery at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.2 **Sampling for lot acceptance.**

4.2.1 *Sampling for inspection.*

4.2.1.1 *Sampling for visual and dimensional inspection.* — A random sample of components shall be taken from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the single sampling inspection requirements of

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table XI in accordance with Military Standard MIL-STD-105.

TABLE XI.—*Sampling for visual and dimensional inspection AQL (approx.) = 1.5 percent defective*

Number of components in inspection lot	Sample size, number to be inspected (reference MIL-STD-105)
40 and under	Inspection Level III
41 to 300	Inspection Level II
301 and over	Inspection Level I

NOTE: The Government inspector may institute tightened or reduced inspection in accordance with Section 9 of Military Standard MIL-STD-105.

4.2.1.2 *Sampling for inspection (sets).*— Sampling for inspection of sets for completeness and container shall be in accordance with Military Standard MIL-STD-105 at inspection level II and AQL = 1.0 percent defective.

4.2.2 *Sampling for tests.* — A random sample of components shall be taken in accordance with table XII, and shall be subjected to the tests specified in 4.4 by or in the presence of the Government inspector to determine conformance with this specification.

TABLE XII.—*Sampling for tests*

Number of components in inspection lot	Number of components in sample	Acceptance number (test failures)	Rejection number (test failures)
Under 110	10	0	1
111 to 180	15	0	1
181 to 500	25	1	2
501 to 800	35	1	2
801 to 1300	50	2	3
1301 to 3200	75	3	4
3201 to 8000	110	4	5
8001 and over	150	6	7

4.3 **Visual and dimensional inspection at place of manufacture.**—Each of the sample components selected in accordance with table XI shall be visually and dimensionally inspected by the Government inspector to determine compliance with this specification. Any component in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective components in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming components. The Government inspector shall again select and examine samples from such rejected lots to determine compliance with this specification.

4.4 Tests at place of manufacture.

4.4.1 *Torque.* — Each component of the sample selected in accordance with table XII shall be tested for rated torque value. The testing equipment shall be made available by the manufacturer. Any component in the sample that exhibits permanent set after application of the torque load shall be rejected, and if the number of defective components in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming components. The Government inspector shall again select and examine samples from such rejected lots to determine compliance with this specification.

4.4.2 *Hardness.*—Ten percent of all sockets selected in accordance with table XII shall be tested to determine compliance with 3.6.1.

5. PREPARATION FOR DELIVERY

5.1 **Preservation.**—Cleaning and preservation shall be in accordance with manufacturer's standard practice.

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5.2 **Packaging.** — Commercial packages will be acceptable.

5.3 **Packing.** — The equipment shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose in such manner as to insure acceptance and safe delivery to the designated point by common or other carrier at the lowest applicable rate of the carrier.

5.4 **Marking.**

5.4.1 *Interior containers.*—Packages and interior containers shall be marked with the name of the material and the name of the contractor, or with a trade-mark of such known character that the source of supply may be readily determined.

5.4.2 *Exterior containers.*—Shipping containers shall be marked with the name of the item, the type and class, and the quantity therein, as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 **Ordering data.** — Procurement documents should specify the following and purchasers should exercise any desired options offered herein. (See 3.2.2, 3.3.3 and 3.3.4.)

(a) Title, number, and date of this specification.

(b) Set number (see 3.8), if sets are required, or the component number of the particular type, class, style, form and size required. (See 1.2 and applicable table.)

(c) Additive finish, if other than specified in 3.2.2. When environmental or coating thickness tests are required, stipulate test specification and place of test.

(d) Whether type I, class 1, style A wrench heads and attachments, type II, class 1, style A, forms A and B should be marked with code. (See 3.3.1.)

(e) Marking requirements for class II, if other than specified in 3.3.3 and 3.3.4.

(f) Whether preparation for domestic shipment-immediate use, domestic shipment and storage, or overseas shipment is required (see sec. 5 and 7.2.3).

6.2 Sets should be supplied only to fill initial outfitting. Minor losses or breakage of components in existing sets should be replaced by individual components of corresponding types and sizes.

6.3 The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of wrenches otherwise conforming to this specification.

6.4 **Transportation description.** — The proper transportation description applicable to these items is:

Wrenches, not otherwise indexed by name, with attachments.
Carload weight 30,000 pounds.
Truckload volume weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specification and standard, of the issues in effect on date of invitation for bids, and special requirements,

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form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standards.*

Military Specification:

MIL-P-15424—Packaging and Packing of Hand Tools; General Specification for Domestic and Overseas Preservation.

Military Standard:

MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Marking for identification.* — Items shall be marked for identification in accordance with Military Standard MIL-STD-130.

7.2.3 *Preparation for delivery.* — Items shall be cleaned, preserved, packaged, packed, and marked in accordance with Military Specification MIL-P-15424. The nomenclature shall be the exact nomenclature for the part or item that has been approved under the contract or order.

MILITARY INTEREST:

Army—O C E Q Sig T
Navy—Sh A *MC* Or
Air Force.

Copies of this specification may be purchased for 15 cents.

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DOCUMENT IDENTIFIER AND TITLE

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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